TrophicTrace

Action Center & Environmental Resources

Complete Technical Reference | YHack 2026

1. System Overview

The TrophicTrace Action Center is the system's primary interface for translating contamination predictions into actionable public health guidance. When a user clicks any river segment or contamination zone on the map, the Action Center slides in from the left side of the screen, presenting a comprehensive risk report for that specific location. The system combines three layers of information: the ML model's contamination predictions and contributing factor analysis, a curated database of 16 environmental resource contacts matched geographically to the user's selected location, and a real-time alert feed aggregated across all monitored segments nationwide.

The Action Center operates as a React component (ActionPanel) that receives the clicked segment's full data object and renders three tabbed views: a detailed Report for the selected location, an All Alerts feed showing every segment above the contamination threshold, and a Resources directory of environmental centers organized by type.

2. Click-Triggered Interaction Model

2.1 Map Click Detection

The map uses Mapbox GL JS with invisible hit-area layers that capture click events on both river LineStrings and contamination zone Polygons. Two invisible layers are registered for click detection: a river-hit-area layer (a transparent LineString layer with a wide stroke width of 20px for easy targeting) and the zone-fill layer (the semi-transparent contamination zone polygons). When the user clicks either layer, Mapbox's queryRenderedFeatures returns the topmost feature at the click point.

The click handler extracts the feature's properties (segment_id/comid, PFAS concentration, coordinates) and performs a lookup in the full data.segments array to retrieve the complete segment object, including species predictions, contributing factors, and the pre-computed nearest environmental center. This full segment object is passed up to the App component via the onSegmentClick callback.

2.2 State Management

The App component manages three independent UI states: hoveredSegment (for the floating tooltip that follows the cursor), selectedSpecies (for the detailed species panel that slides in from the right), and clickedSegment (for the Action Center). These states are mutually exclusive in the following way: when a segment is clicked, the hover tooltip is dismissed; when the Action Center is open, hover tooltips are suppressed; when a species is selected from the hover tooltip, the Action Center closes. The conditional rendering logic in the App's JSX ensures only one panel type is visible at a time:

```
// Hover tooltip: only when no panels are open
```

```
{hoveredSegment && !selectedSpecies && !clickedSegment && (
  <FloatingIsland ... />
)}


// Species detail: from FloatingIsland click
{selectedSpecies && (
  <DetailPanel ... />
)}


// Action Center: from map click
{clickedSegment && !selectedSpecies && (
  <ActionPanel segment={clickedSegment} data={data} ... />
)}
```

The clickedSegment state is set by handleSegmentClick and cleared by the Action Center's close button. A useEffect inside ActionPanel resets the active tab to "Report" whenever a new segment is clicked, so the user always sees the risk report first.

3. Report Tab: Risk Assessment

3.1 Urgency Classification

Each segment is classified into one of three urgency tiers based on its predicted water-column PFAS concentration. These thresholds are calibrated against EPA health advisory levels and state fish consumption advisory triggers:

Tier	PFAS Threshold	Recommended Action
CRITICAL	≥ 500 ng/L	Do not consume fish. Contact state PFAS response team immediately.
WARNING	≥ 200 ng/L	Limit consumption to ≤1 serving/month. Check state advisories.
CAUTION	50–199 ng/L	Monitor advisories. Recreational anglers: 2–4 servings/month max.
SAFE	< 50 ng/L	Currently within safe limits. Continue monitoring for changes.

The urgency tier drives the entire visual treatment of the Action Center: the header badge color (red/amber/green), the recommendation card's border and background tint, and the shield icon color. This creates an immediate, unmistakable visual signal of risk severity.

3.2 Most Affected Species

The report identifies the single most contaminated fish species at the clicked location. This is computed by iterating the segment's species array and selecting the one with the highest tissue_total_pfas_ng_g value. The species card displays the common name, the tissue concentration in ng/g, and a consumption advisory (either "Do not consume" for unsafe species, or the maximum safe servings per month for limited-safety species). The tissue concentration is the sum across all 6 PFAS congeners predicted by the Physics-Informed Neural Network.

3.3 Biggest Contributing Factors

This is one of the most technically significant features of the Action Center. For each segment, the XGBoost model's per-prediction feature contributions are extracted and displayed as a ranked bar chart showing the top 5 factors driving contamination at that specific location.

How Feature Contributions Are Computed

When the XGBoost model makes a prediction for a segment, it internally traverses 250 decision trees. At each tree split, the sample goes left or right based on its feature value, and each split contributes a fractional amount to the final prediction. The built-in tree SHAP method (booster.predict with pred_contribs=True) computes, for each of the 29 input features, its additive contribution to the log-scale prediction. For a given segment i:

```
contributions = booster.predict(dmatrix, pred_contribs=True)

# Shape: (n_samples, 30) — 29 features + 1 bias term

# contributions[i][j] = feature j's additive effect on log(pfas)



abs_contributions = |contributions[i][0:29]|

normalized = abs_contributions / sum(abs_contributions)

top_5 = sorted(normalized, descending)[:5]
```

The absolute values are taken because both positive and negative contributions indicate influence. Normalization to sum to 1.0 converts raw SHAP values into interpretable percentages. The top 5 are retained per segment and stored as the top_contributing_features array in the segment data.

Feature Label Mapping

Raw feature names from the model (like nearest_pfas_facility_km or upstream_npdes_pfas_count) are mapped to human-readable labels for display. The complete mapping used in the Action Center is:

Model Feature Name	Display Label
nearest_pfas_facility_km	Proximity to PFAS facility
upstream_npdes_pfas_count	Upstream PFAS dischargers
low_flow_7q10_m3s	Low-flow conditions
pct_urban	Urban land use
dissolved_organic_carbon_mgl	Dissolved organic carbon
mean_annual_flow_m3s	River flow rate
afff_site_nearby	Nearby AFFF site
wwtp_upstream	Upstream wastewater plant
pfas_industry_density	PFAS industry density
stream_order	Stream order
baseflow_index	Baseflow index
pct_agriculture	Agricultural land use
pct_impervious	Impervious surface
population_density	Population density

Each factor is rendered as a labeled progress bar. The bars are color-coded by rank: the top contributor is red (#EF4444), the second is orange (#F97316), and the remaining three are amber (#F59E0B). This visual hierarchy draws the eye to the most important driver.

3.4 Species Roster

Below the contributing factors, the report lists all fish species predicted at the segment, sorted by tissue contamination from highest to lowest. Each species row shows a colored status dot (red for unsafe, amber for limited, green for safe), the common name, and the tissue PFAS concentration in ng/g. The species data comes from the PINN model's predictions across 8 species and 6 congeners, summed per species.

4. Environmental Centers Database

4.1 Data Source and Structure

The system includes a curated database of 16 environmental resource contacts covering three tiers of PFAS response infrastructure. These were compiled from official EPA regional office directories, state environmental agency PFAS program pages, and national environmental hotline registries.

Type	Count	Description

EPA Regional Offices	8	EPA Regions 1–5, 7, 8, and 9 covering the 10 monitored states. Each has enforcement authority over PFAS dischargers.
State PFAS Programs	6	State-level PFAS response teams in MI, NC, VT, NH, NJ, PA. These handle fish consumption advisories and local testing.
National Hotlines	2	EPA Safe Drinking Water Hotline and ATSDR PFAS Information Line. Available nationwide.

Each center record contains: name (official office/program name), lat and lng (geographic coordinates of the office), phone (direct phone number), url (official PFAS resource webpage), states (array of state abbreviations the office covers), and type (epa_regional, state_pfas, or national_hotline).

4.2 Geographic Matching Algorithm

The core innovation in the environmental center system is automatic geographic matching. When the data is prepared for the frontend, every segment is matched to its nearest environmental center using the Haversine distance formula. This ensures that when a user clicks any location on the map, they immediately see the most relevant contact without needing to know which EPA region or state program covers that area.

Haversine Distance Computation

The Haversine formula computes the great-circle distance between two points on Earth's surface, accounting for the planet's curvature. Given two points (lat1, lon1) and (lat2, lon2) in radians:

```
function haversine(lat1, lon1, lat2, lon2) {

  const R = 6371  // Earth radius in km

  const dLat = toRadians(lat2 - lat1)

  const dLon = toRadians(lon2 - lon1)

  const a = sin(dLat/2)^2 +

            cos(toRadians(lat1)) * cos(toRadians(lat2)) *

            sin(dLon/2)^2

  const c = 2 * atan2(sqrt(a), sqrt(1-a))

  return R * c  // distance in kilometers

}
```

For each of the 300 segments, the algorithm iterates through all 16 environmental centers and selects the one with the minimum Haversine distance. The matched center is embedded directly into the segment's data object as nearest_env_center, along with the computed

distance_km. This pre-computation happens at build time during the data integration step, so the frontend performs zero distance calculations at runtime.

Matching Priority

The current implementation uses pure geographic proximity, which means a segment near the border of two EPA regions will be matched to whichever office is physically closer. In practice, this works well because EPA regional offices and state PFAS programs are located near the populations they serve. National hotlines (which have coordinates set to Washington, D.C.) serve as fallbacks for segments far from any regional office. Future versions could incorporate a state-boundary check: if the segment's state matches a state PFAS program's coverage area, prefer that program even if an EPA regional office is geographically closer.

4.3 Frontend Display

When the Action Center's Report tab is open, the nearest environmental center is displayed as a blue-accented card with the center's name, a clickable phone link (tel: protocol for mobile tapping), a website link, and the distance in kilometers. On the Resources tab, all 16 centers are listed grouped by type: National Hotlines first (highest urgency), then State PFAS Programs (most locally relevant), then EPA Regional Offices (for escalation). Each group has color-coded left borders: purple for national hotlines, blue for state programs, and cyan for EPA offices.

5. Alert System

5.1 Alert Generation

The alert feed is generated client-side using React's useMemo hook to efficiently process all 300 segments. The algorithm filters segments where predicted_water_pfas_ng_l >= 100 or risk_level === 'high', then classifies each into the three urgency tiers (critical/warning/caution using the same thresholds as the Report tab). For each alert, the system identifies the worst-affected species (highest tissue PFAS) and attaches the pre-computed nearest environmental center. Alerts are sorted by PFAS concentration in descending order, so the most dangerous locations appear first.

```
const allAlerts = useMemo(() => {

  return data.segments

    .filter(seg => seg.predicted_water_pfas_ng_l >= 100

                || seg.risk_level === 'high')

    .map(seg => {

      const pfas = seg.predicted_water_pfas_ng_l

      const urgency = pfas >= 500 ? 'critical'

                    : pfas >= 200 ? 'warning'
```

```
                    : 'caution'

      const worstSpecies = seg.species?.reduce((a, b) =>

        b.tissue_total_pfas_ng_g > a.tissue_total_pfas_ng_g ? b : a)

      return { segment: seg, pfas, urgency, worstSpecies }

    })

    .sort((a, b) => b.pfas - a.pfas)

}, [data])
```

The useMemo dependency on data ensures alerts are only recomputed when the underlying dataset changes, not on every render cycle. With 300 segments, the filter-map-sort operation completes in under 1ms.

5.2 Alert Cards

Each alert is rendered as a compact card with the urgency tier badge, PFAS concentration, segment name, and worst species with its tissue level. Cards are color-coded: red background tint for critical, amber for warning, green for caution. A colored left border reinforces the urgency signal. The first 5 alerts are shown by default; a “Show all N alerts” button expands the list.

5.3 Map Fly-To Navigation

Clicking an alert card triggers the onAlertClick callback, which performs two actions. First, it calls mapRef.current.flyTo() with the alert’s segment coordinates, zooming to level 8 with a 1.5-second animation duration. This smoothly pans and zooms the map to center on the alert’s location. Second, it sets clickedSegment to the alert’s segment, which updates the Action Center’s Report tab to show that segment’s details. The effect is seamless: the user clicks an alert, the map flies to the location, and the report updates to show the local risk assessment. The implementation:

```
const handleAlertClick = (segment) => {

  if (mapRef.current && segment) {

    const lat = segment.lat ?? segment.latitude

    const lng = segment.lng ?? segment.longitude

    mapRef.current.flyTo({

      center: [lng, lat],

      zoom: 8,

      duration: 1500

    })
```

```
  }

  setClickedSegment(segment)

}
```

Note the nullish coalescing for lat/lng fields, which handles the data format inconsistency between segments that use lat/lng versus latitude/longitude field names.

6. Map Visualization Integration

6.1 Environmental Center Markers

Environmental centers are rendered on the Mapbox map as diamond-shaped markers (45° rotated squares) at each center's geographic coordinates. The markers are color-coded by type: cyan (#06B6D4) for EPA Regional, blue (#3B82F6) for State PFAS, and purple (#8B5CF6) for National Hotline. They are added as a symbol layer using a Mapbox expression that maps the type property to a color, with a rotation of 45 degrees for the diamond appearance. The markers are visible at all zoom levels, providing persistent geographic context.

6.2 Contamination Zone Interaction

Both LineString features (rivers/tributaries) and Polygon features (contamination zones, source areas, ocean patches) are clickable. The map separates these into two GeoJSON sources at load time by filtering the feature type. Rivers get line layers with glow effects; polygons get semi-transparent fill layers with dashed outlines. Both register click handlers that invoke the same onSegmentClick callback, ensuring the Action Center works identically whether the user clicks a river line or a contamination zone polygon.

6.3 Visual Design System

The Action Center uses a dark, translucent design that overlays the map without obscuring it. The panel background is rgba(25, 25, 25, 0.94) with a 16px backdrop blur, creating a frosted glass effect. It slides in from the left with a CSS keyframe animation (250ms ease-out translateX). All text uses CSS custom properties (--text-primary, --text-secondary, --text-tertiary) that are defined in the global stylesheet and tuned for readability against the dark map. The three-tab navigation uses bottom-border highlighting on the active tab, matching the minimal aesthetic of the overall dark-v11 Mapbox basemap theme.

7. Exposure Disparity Analysis

7.1 Hazard Quotient Comparison

The Detail Panel (opened when clicking a specific fish species) includes an exposure disparity visualization that highlights the disproportionate risk faced by subsistence fishers compared to recreational anglers. Two horizontal bars represent the Hazard Quotient (HQ) for each population:

Recreational angler HQ uses a consumption rate of 17 g/day (approximately 2 meals per month). Subsistence fisher HQ uses 142.4 g/day (approximately 1 meal per day), representing

communities that rely on locally caught fish as a primary protein source. The subsistence rate is 8.4x higher, which directly multiplies the tissue-based hazard. For a fish with 50 ng/g total PFAS, the recreational HQ might be 0.8 (below the EPA safety threshold of 1.0) while the subsistence HQ is 6.7 (far exceeding it).

The bars are drawn relative to a shared maximum scale, with a dashed vertical line at the HQ = 1.0 threshold. Colors shift from green (HQ < 0.5) to amber (0.5–1.0) to red (> 1.0), making it visually obvious when a population exceeds safe exposure levels.

7.2 Demographic Context

The exposure analysis is enriched with nearby demographic data. The system includes 5 demographic zones with census-derived statistics: community name, median household income, percentage of households relying on locally caught fish (subsistence_pct), predominant nearby fishing type, and geographic coordinates. A nearest-demographic lookup (using Euclidean distance with a 0.5-degree threshold) matches the current segment to the closest demographic zone and displays a contextual statement like “Near Fayetteville Communities (median income $38,200), 12% of households rely on locally caught fish.” This surfaces environmental justice concerns directly within the risk assessment.

8. Backend API Endpoints

8.1 Alert Generation API

The backend server exposes GET /api/alerts, which performs the same alert generation logic server-side. It iterates all 300 segments from the in-memory national results, filters for predicted_water_pfas_ng_l >= 100, classifies urgency, identifies the worst species per segment, and returns a JSON array sorted by severity. Response fields include segment_id, name, latitude, longitude, pfas_ng_l, urgency, recommended_action, worst_species (name and tissue concentration), and nearest_center (name, phone, distance). This endpoint enables future integrations: mobile push notifications, email alerts, or third-party dashboards.

8.2 Environmental Centers API

GET /api/environmental-centers returns the full list of 16 centers with all metadata. GET /api/nearest-center?lat=X&lng=Y accepts coordinates and returns the single closest center with distance, using the same Haversine computation as the build-time matching. This endpoint supports real-time queries from mobile clients or external applications that cannot use the pre-computed matches.

8.3 Live Water Data Integration

GET /api/live-water-data fetches real-time water quality from the USGS Instantaneous Values API for temperature, dissolved oxygen, and pH across 8 monitored states. Results are cached for 15 minutes in a global dictionary. When live data is available for a segment (matched by geographic proximity), the Action Center could display a “Live Readings” section showing current water conditions alongside the model’s PFAS predictions. This bridge between modeled and measured data increases user confidence in the system’s assessments.